

Homework 2

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- This is a block cipher because it permutes the bits of a length n blocks.

$$C_1 = m_1 \oplus K_1 = 101 \oplus 111 = 010$$

$$C_2 = m_1 \oplus m_2 \oplus m_3 = 101 \oplus 111 \oplus 010 = 000$$

$$C_3 = m_3 \oplus K_2 = 010 \oplus 011 = 001$$

$$C_4 = m_4 \oplus K_3 = 110 \oplus 101 = 011$$

$$C = 010 \ 000 \ 001 \ 011$$

$$m_1 = C_1 \oplus K_1 = 001 \oplus 111 = 110$$

$$m_2 = m_1 \oplus C_2 \oplus m_3 = 110 \oplus 100 \oplus 101 = 111$$

$$m_3 = C_3 \oplus K_2 = 110 \oplus 011 = 101$$

$$m_4 = C_4 \oplus K_3 = 010 \oplus 101 = 111$$

$$m = 110 \ 111 \ 101 \ 111$$

- Only C_2 can be computed because there is no key involved in it.

$$C_2 = m_1 \oplus m_2 \oplus m_3 = 111 \oplus 111 \oplus 111 = 111$$

All the other blocks (C_1 , C_3 and C_4) can not be computed.

$$C_2 = m_1 \oplus m_2 \oplus m_3 = 111 \oplus 000 \oplus 111 = 000$$

• This scheme does not have perfect secrecy since the key size is smaller than the message size (the key size is $3n$, and the message size is $4n$).

• In the semantic security game:

- $P(w_0) = 1$. In Experiment 0, we have $c_2 = 1 \dots 1$ so the adversary returns 1.
- $P(w_1) = 0$. In Experiment 1, we have $c_2 = 0 \dots 0$ so the adversary returns 0.

$$A = |P(w_0) - P(w_1)| = |1 - 0| = 1$$